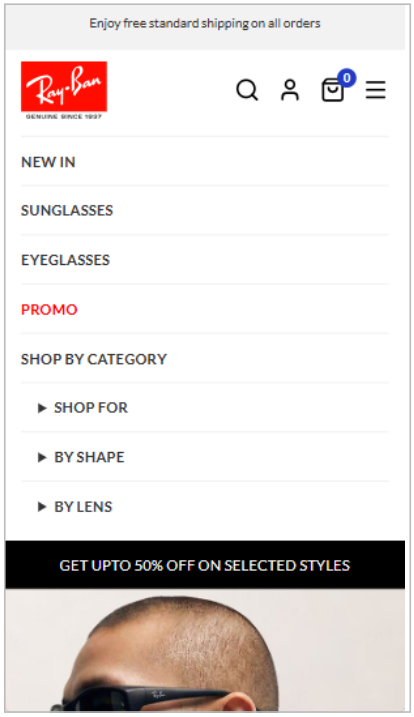
Web Technologies (CSC336) - Midterm Lab

COMSATS University Islamabad, Lahore Campus - FA26 BCS B Lab Mid Term

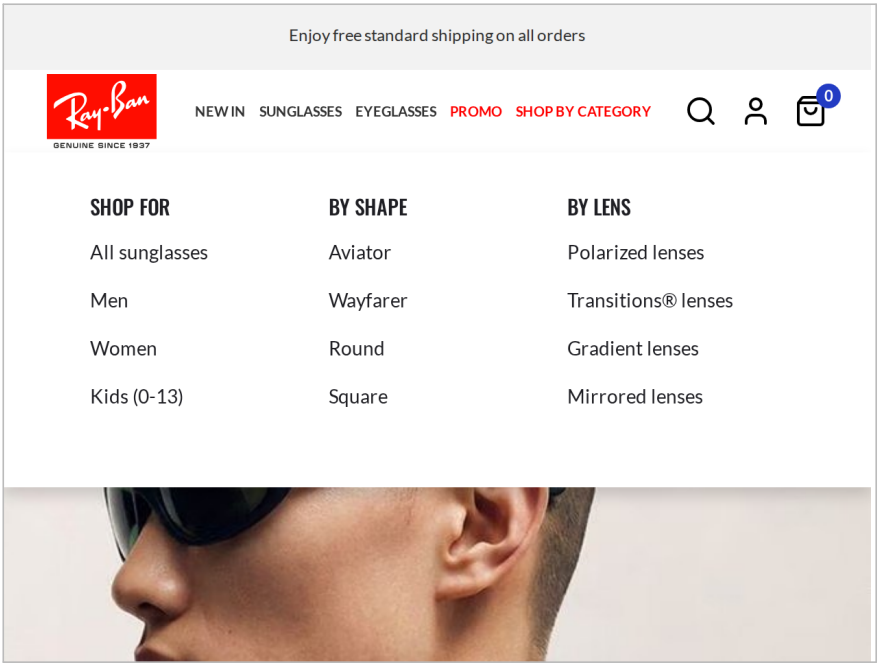
Submitted By	Fatima Khan
Registration No	FA24-BCS-033
Submitted To	Sir Usman Akram
Date of Submission	29th October 2026
My numbers	N = 33, BP = 633px, H = 231 (hsl(231, 70%, 45%)), K = 3

Output

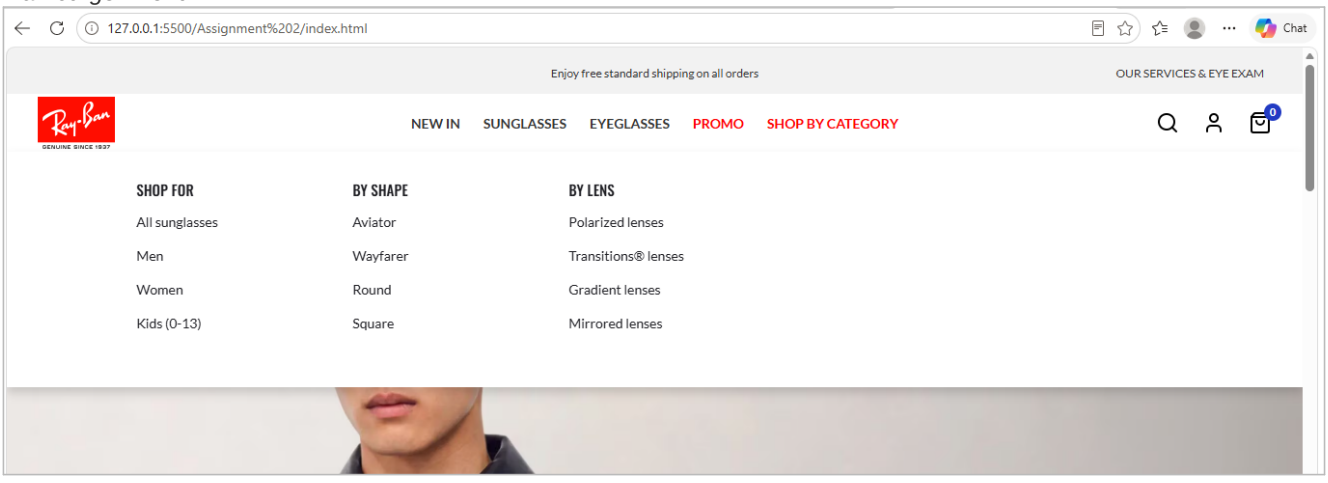
Task 1 - Mega-menu



Phone (375px): "Shop by Category" as three closed accordions inside the hamburger menu

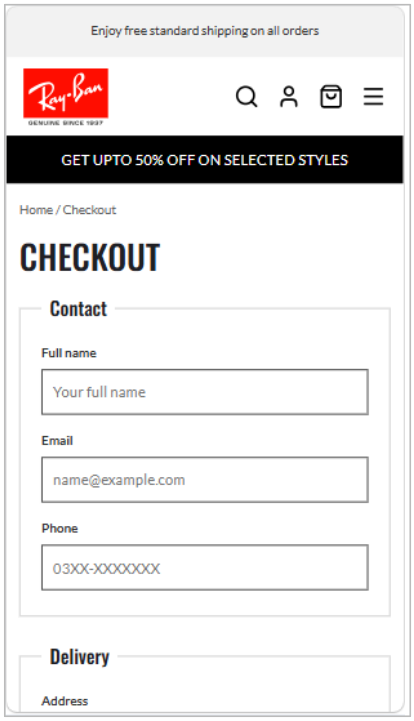


BP (633px): mega-menu open under the navbar, 3 columns

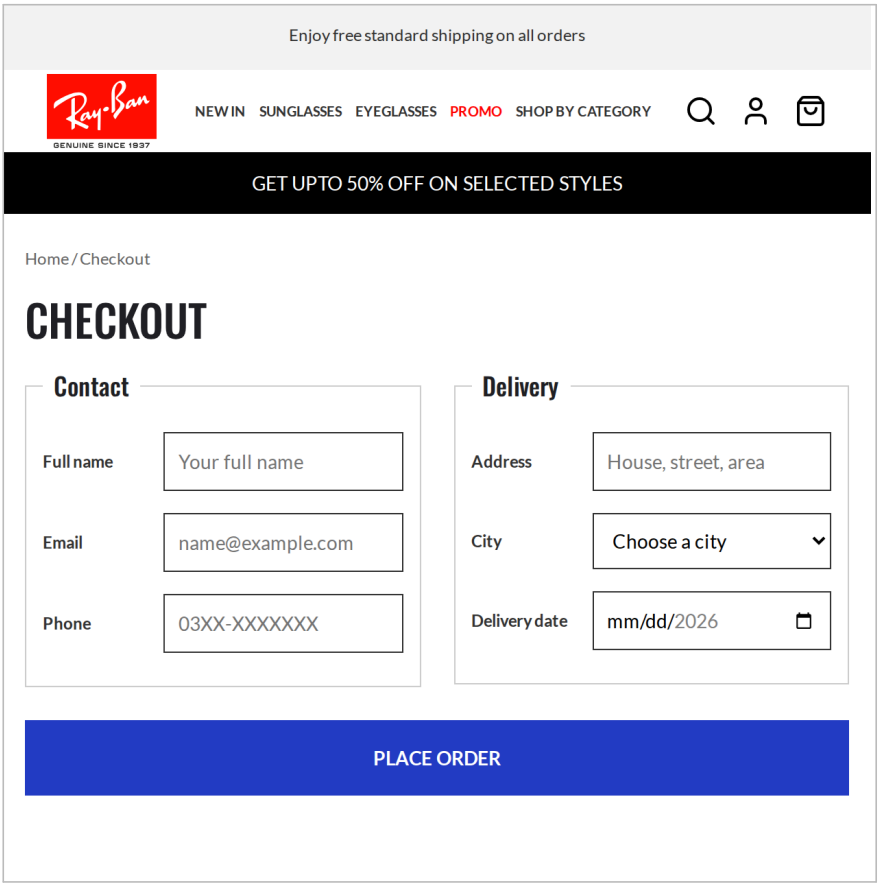


Desktop (1280px): mega-menu open, 3 columns of 4 links with a heading each

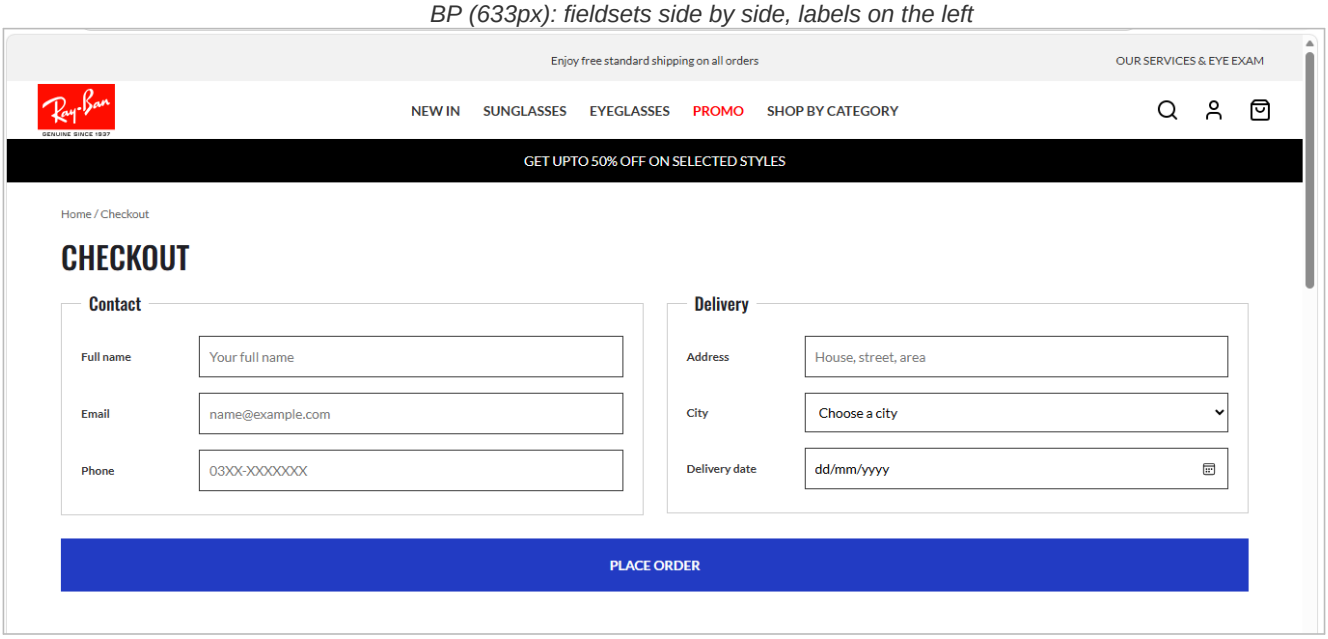
Task 2 - Checkout form



Phone (375px): one column, labels above inputs

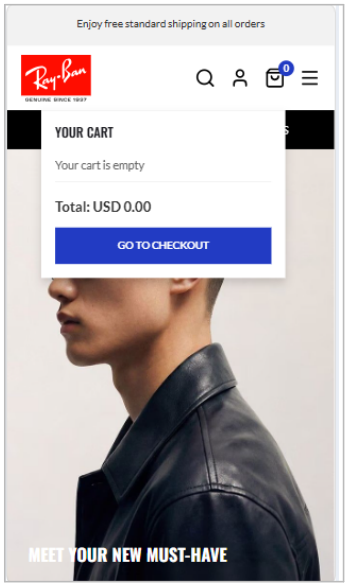


BP (633px): fieldsets side by side, labels on the left

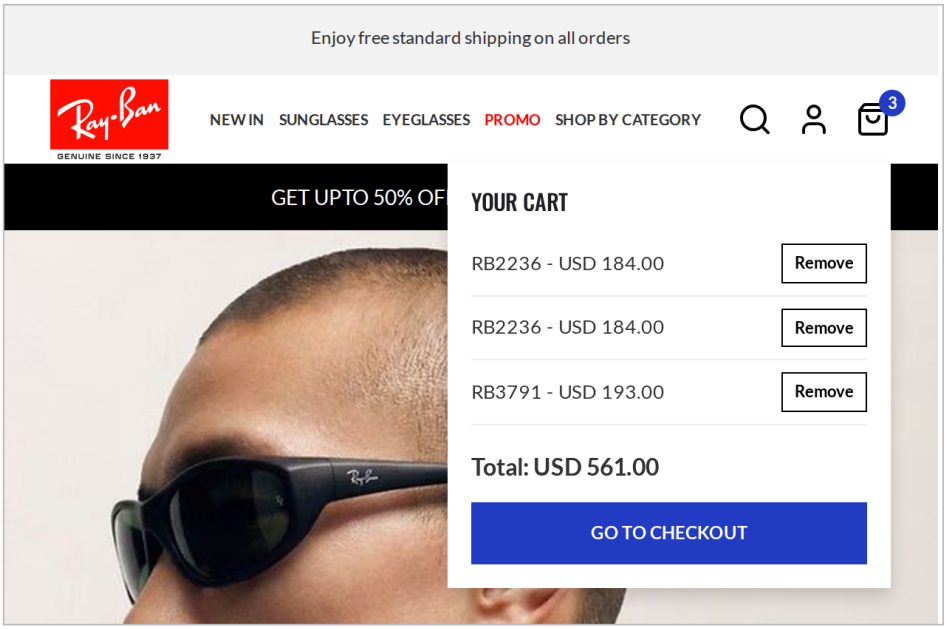


Desktop (1280px): fieldsets side by side, full-width submit button

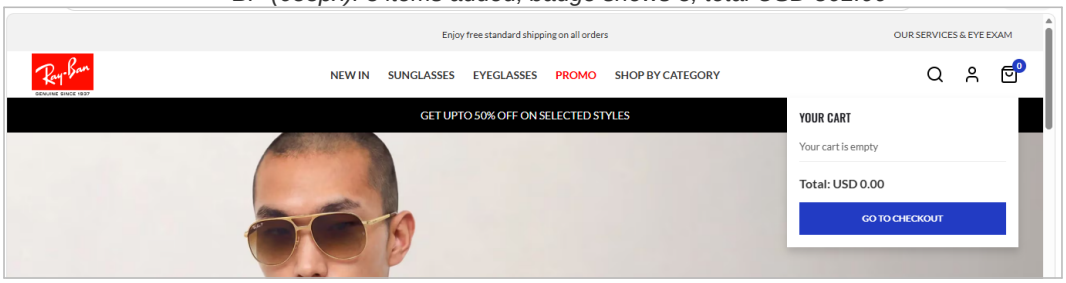
Task 3 - Cart badge and mini cart



Phone (375px): mini-cart panel open under the cart icon

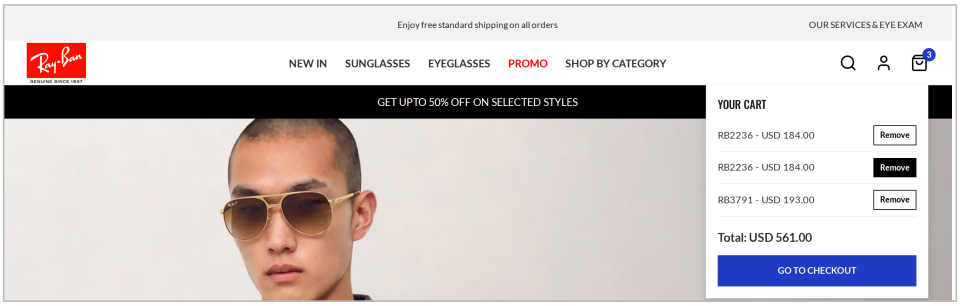


BP (633px): 3 items added, badge shows 3, total USD 561.00

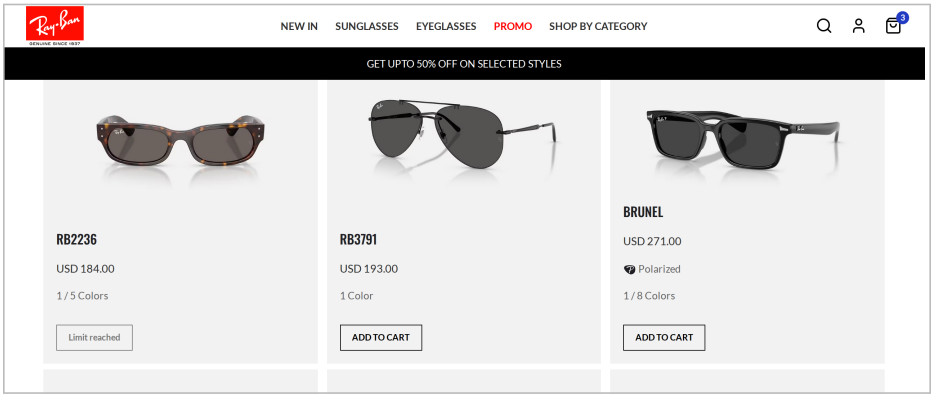


Desktop (1280px): mini-cart panel open under the cart icon

Task 3 - After using the cart



After clicking Remove: the list, the total (USD 561.00) and the badge (3) all updated at once



Product at the limit: after 3 adds, RB2236 shows "Limit reached" and its button is disabled

Code

Task 1 - Mega-menu

index.html, sunglasses.html, product.html (link added after styles.css, menu div changed to nav):

```
<link rel="stylesheet" href="styles.css" />
<link rel="stylesheet" href="midterm.css" />

<nav class="nav-links collapse navbar-collapse" id="navLinks">
```

index.html, sunglasses.html, product.html (new menu item added at the end of the nav):

```
  <div class="menu-item category-item">
    <a href="#">SHOP BY CATEGORY</a>
    <div class="big-menu">
      <div class="menu-columns">
        <div class="menu-column">
          <h4>SHOP FOR</h4>
          <ul>
            <li><a href="sunglasses.html">All sunglasses</a></li>
            <li><a href="#">Men</a></li>
            <li><a href="#">Women</a></li>
            <li><a href="#">Kids (0-13)</a></li>
          </ul>
        </div>
        <div class="menu-column">
          <h4>BY SHAPE</h4>
          <ul>
            <li><a href="#">Aviator</a></li>
            <li><a href="#">Wayfarer</a></li>
            <li><a href="#">Round</a></li>
            <li><a href="#">Square</a></li>
          </ul>
        </div>
        <div class="menu-column">
          <h4>BY LENS</h4>
          <ul>
            <li><a href="#">Polarized lenses</a></li>
            <li><a href="#">Transitions&reg; lenses</a></li>
            <li><a href="#">Gradient lenses</a></li>
            <li><a href="#">Mirrored lenses</a></li>
          </ul>
        </div>
      </div>
    </div>
    <div class="category-accordion">
      <details>
        <summary>SHOP FOR</summary>
        <ul>
          <li><a href="sunglasses.html">All sunglasses</a></li>
          <li><a href="#">Men</a></li>
          <li><a href="#">Women</a></li>
          <li><a href="#">Kids (0-13)</a></li>
        </ul>
      </details>
      <details>
        <summary>BY SHAPE</summary>
        <ul>
          <li><a href="#">Aviator</a></li>
          <li><a href="#">Wayfarer</a></li>
          <li><a href="#">Round</a></li>
          <li><a href="#">Square</a></li>
        </ul>
      </details>
      <details>
        <summary>BY LENS</summary>
        <ul>
          <li><a href="#">Polarized lenses</a></li>
          <li><a href="#">Transitions&reg; lenses</a></li>
          <li><a href="#">Gradient lenses</a></li>
          <li><a href="#">Mirrored lenses</a></li>
        </ul>
      </details>
    </div>
  </div>
</nav>
```

midterm.css (new file):

```
.menu-column ul,
.category-accordion ul {
  list-style: none;
}

.menu-column ul {
```

```
  padding: 0;
}

.category-accordion {
  display: none;
}

.category-item .big-menu {
  z-index: 2000;
}

.category-item:focus-within .big-menu {
  display: block;
}

@media (min-width: 633px) {
  .menu-icon {
    display: none;
  }

  .nav-links.collapse:not(.show) {
    display: flex;
  }
}

@media (min-width: 633px) and (max-width: 767px) {
  .navigation-bar {
    height: 60px;
    min-height: 0;
    padding: 0 32px;
  }

  .nav-links {
    position: absolute;
    flex-direction: row;
    align-items: center;
    pointer-events: none;
    width: auto;
    order: 0;
    padding: 0 140px 0 120px;
  }

  .menu-item {
    height: 60px;
    padding: 0 5px;
  }

  .menu-item > a {
    display: inline;
    width: auto;
    padding: 0;
    border-top: 0;
    font-size: 10px;
  }

  .menu-item:hover .big-menu,
  .category-item:focus-within .big-menu {
    display: block !important;
  }
}

@media (max-width: 632px) {
  .category-item {
    flex-direction: column;
    align-items: stretch;
  }

  .category-accordion {
    display: block;
  }

  .category-accordion summary {
    font-size: 14px;
    font-weight: bold;
    padding: 12px 0 12px 16px;
    border-top: 1px solid #eee;
    cursor: pointer;
  }

  .category-accordion ul {
    padding: 0 0 8px 32px;
  }

  .category-accordion li a {
    display: block;
    font-size: 14px;
    padding: 6px 0;
  }
```

```
}
```

Task 2 - Checkout form

index.html, sunglasses.html, product.html (cart button changed to a link; on index.html and sunglasses.html it is wrapped by Task 3 below):

```
<a class="cart-icon" href="checkout.html" aria-label="Cart">
  <img src="assets/shopping-bag.svg" alt="Cart icon" />
</a>
```

checkout.html (new file, form part of the page):

```
<div class="checkout-page">
  <p class="breadcrumb-text"><a href="index.html">Home</a> / Checkout</p>
  <h1 class="page-title">CHECKOUT</h1>
  <form class="checkout-form" action="#" method="get">
    <fieldset>
      <legend>Contact</legend>
      <div class="field">
        <label for="fullName">Full name</label>
        <input type="text" id="fullName" name="fullName" placeholder="Your full name" required />
        <span class="error-msg">Please enter your full name.</span>
      </div>
      <div class="field">
        <label for="email">Email</label>
        <input type="email" id="email" name="email" placeholder="name@example.com" required />
        <span class="error-msg">Enter a valid email address.</span>
      </div>
      <div class="field">
        <label for="phone">Phone</label>
        <input type="tel" id="phone" name="phone" placeholder="03XX-XXXXXXX" pattern="03[0-9]{2}-[0-9]{7}"
            title="Use the format 03XX-XXXXXXX" required />
        <span class="error-msg">Phone must look like 0300-1234567.</span>
      </div>
    </fieldset>
    <fieldset>
      <legend>Delivery</legend>
      <div class="field">
        <label for="address">Address</label>
        <input type="text" id="address" name="address" placeholder="House, street, area" required />
        <span class="error-msg">Please enter your address.</span>
      </div>
      <div class="field">
        <label for="city">City</label>
        <select id="city" name="city" required>
          <option value="">Choose a city</option>
          <option value="lahore">Lahore</option>
          <option value="karachi">Karachi</option>
          <option value="islamabad">Islamabad</option>
          <option value="rawalpindi">Rawalpindi</option>
          <option value="faisalabad">Faisalabad</option>
          <option value="multan">Multan</option>
        </select>
        <span class="error-msg">Please choose a city.</span>
      </div>
      <div class="field">
        <label for="deliveryDate">Delivery date</label>
        <input type="date" id="deliveryDate" name="deliveryDate" min="2026-10-29" max="2026-12-01" required />
        <span class="error-msg">Pick a date from 29 Oct 2026 to 1 Dec 2026.</span>
      </div>
    </fieldset>
    <button class="checkout-btn" type="submit">PLACE ORDER</button>
  </form>
</div>
```

midterm.css (rules added for Task 2):

```
.checkout-page {
  padding: 24px 56px 56px 56px;
}

.checkout-form {
  display: grid;
  grid-template-columns: 1fr;
  gap: 24px;
  margin-top: 16px;
}

.checkout-form fieldset {
  border: 1px solid #ccc;
  padding: 8px 20px 8px 20px;
  min-width: 0;
}

.checkout-form legend {
```

```
  float: none;
  width: auto;
  padding: 0 8px;
  margin-bottom: 0;
  font-family: "Oswald", sans-serif;
  font-weight: 500;
  font-size: 18px;
  color: #1f1f24;
}

.field {
  display: grid;
  grid-template-columns: 1fr;
  align-items: center;
  margin: 12px 0 16px 0;
}

.field label {
  font-size: 12px;
  font-weight: bold;
  margin-bottom: 6px;
}

.field input,
.field select {
  width: 100%;
  padding: 10px;
  border: 1px solid #333;
  font-size: 14px;
  background-color: white;
}

.field input:focus,
.field select:focus {
  outline: 2px solid hsl(231, 70%, 45%);
  outline-offset: 0;
}

.error-msg {
  display: none;
  color: red;
  font-size: 12px;
  margin-top: 4px;
}

.field input:user-invalid,
.field select:user-invalid {
  border-color: red;
}

.field input:user-invalid ~ .error-msg,
.field select:user-invalid ~ .error-msg {
  display: block;
}

.checkout-btn {
  grid-column: 1 / -1;
  width: 100%;
  padding: 16px;
  background-color: hsl(231, 70%, 45%);
  color: white;
  border: 1px solid hsl(231, 70%, 45%);
  font-size: 14px;
  font-weight: bold;
}

.checkout-btn:hover {
  background-color: white;
  color: hsl(231, 70%, 45%);
}

@media (min-width: 633px) {
  .checkout-form {
    grid-template-columns: 1fr 1fr;
    align-items: start;
  }

  .field {
    grid-template-columns: 110px 1fr;
    column-gap: 12px;
  }

  .field label {
    margin-bottom: 0;
  }

  .error-msg {
    grid-column: 2;
```

```
  }
}

@media (max-width: 632px) {
  .checkout-page {
    padding: 16px 12px 32px 12px;
  }
}
```

midterm.css (rules added at the end so the form fits at BP, 633px to 900px):

```
@media (min-width: 633px) and (max-width: 900px) {
  .checkout-page {
    padding: 24px 16px 40px 16px;
  }

  .checkout-form fieldset {
    padding: 8px 12px 8px 12px;
  }

  .field {
    grid-template-columns: 80px 1fr;
    column-gap: 8px;
  }
}
```

Task 3 - Cart badge and mini cart

index.html, sunglasses.html (cart icon wrapped with badge and mini cart):

```
<div class="cart-wrap">
  <a class="cart-icon" href="checkout.html" aria-label="Cart">
    <img src="assets/shopping-bag.svg" alt="Cart icon" />
    <span class="cart-badge" id="cartBadge">0</span>
  </a>
  <div class="mini-cart" id="miniCart">
    <h4>YOUR CART</h4>
    <ul class="mini-list" id="miniList"></ul>
    <p class="mini-total">Total: <span id="miniTotal">USD 0.00</span></p>
    <a class="mini-checkout" href="checkout.html">GO TO CHECKOUT</a>
  </div>
</div>
```

index.html, sunglasses.html (script added after the Bootstrap script):

```
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
<script src="script.js"></script>
```

script.js (new file):

```
const maxTimes = 3;
let cart = [];

const badge = document.getElementById("cartBadge");
const cartIcon = document.querySelector(".cart-icon");
const miniCart = document.getElementById("miniCart");
const miniList = document.getElementById("miniList");
const miniTotal = document.getElementById("miniTotal");
const addButtons = document.querySelectorAll(".add-btn");

function countOf(name) {
  return cart.filter(function (item) {
    return item.name === name;
  }).length;
}

function showCart() {
  badge.textContent = cart.length;

  miniList.innerHTML = "";
  if (cart.length === 0) {
    miniList.innerHTML = '<li class="mini-empty">Your cart is empty</li>';
  }
  cart.forEach(function (item, index) {
    const li = document.createElement("li");
    li.innerHTML =
      "<span>" + item.name + " - USD " + item.price.toFixed(2) + "</span>" +
      '<button type="button" class="remove-btn" data-index="' + index + '">Remove</button>';
    miniList.appendChild(li);
  });

  const total = cart.reduce(function (sum, item) {
    return sum + item.price;
  }, 0);
  miniTotal.textContent = "USD " + total.toFixed(2);
  addButtons.forEach(function (btn) {
```

```
    const name = btn.closest(".big-card").querySelector(".card-name").textContent.trim();
    if (countOf(name) >= maxTimes) {
      btn.textContent = "Limit reached";
      btn.disabled = true;
    } else {
      btn.textContent = "ADD TO CART";
      btn.disabled = false;
    }
  });
}

addButtons.forEach(function (btn) {
  btn.addEventListener("click", function () {
    const card = btn.closest(".big-card");
    const name = card.querySelector(".card-name").textContent.trim();
    const price = parseFloat(card.querySelector(".card-price").textContent.replace("USD", ""));
    cart.push({ name: name, price: price });
    showCart();
  });
});

miniList.addEventListener("click", function (e) {
  if (e.target.classList.contains("remove-btn")) {
    cart.splice(Number(e.target.dataset.index), 1);
    showCart();
  }
});

cartIcon.addEventListener("click", function (e) {
  e.preventDefault();
  miniCart.classList.toggle("open");
});

showCart();
```

midterm.css (rules added for Task 3):

```
.cart-icon {
  position: relative;
}

.cart-badge {
  position: absolute;
  top: -8px;
  right: -10px;
  min-width: 18px;
  height: 18px;
  padding: 0 4px;
  border-radius: 9px;
  background-color: hsl(231, 70%, 45%);
  color: white;
  font-size: 11px;
  font-weight: bold;
  line-height: 18px;
  text-align: center;
}

.cart-wrap {
  display: flex;
}

.mini-cart {
  display: none;
  position: absolute;
  top: 100%;
  right: 32px;
  width: 300px;
  padding: 16px;
  background-color: white;
  box-shadow: 0 6px 10px rgba(0, 0, 0, 0.15);
  z-index: 2000;
}

.mini-cart.open {
  display: block;
}

.mini-cart h4 {
  font-family: "Oswald", sans-serif;
  font-weight: 500;
  font-size: 15px;
  color: #1f1f24;
  margin-bottom: 12px;
}

.mini-list {
  list-style: none;
  padding: 0;
```

```
  max-height: 240px;
  overflow-y: auto;
}

.mini-list li {
  display: flex;
  justify-content: space-between;
  align-items: center;
  gap: 8px;
  padding: 8px 0;
  border-bottom: 1px solid #eee;
  font-size: 13px;
}

.mini-empty {
  color: #666;
}

.remove-btn {
  background-color: transparent;
  border: 1px solid black;
  font-size: 11px;
  font-weight: bold;
  padding: 4px 8px;
}

.remove-btn:hover {
  background-color: black;
  color: white;
}

.mini-total {
  font-weight: bold;
  margin: 12px 0;
}

.mini-checkout {
  display: block;
  text-align: center;
  background-color: hsl(231, 70%, 45%);
  color: white;
  font-size: 12px;
  font-weight: bold;
  padding: 12px;
}

.add-btn:disabled {
  opacity: 0.5;
  cursor: not-allowed;
}

.add-btn:disabled:hover {
  background-color: transparent;
  color: black;
}

@media (max-width: 632px) {
  .mini-cart {
    right: 56px;
    width: 280px;
  }
}
```